



Diagnostic value of circulating lncRNAs HOTAIR, ANRIL, and MEG3 in oral squamous cell carcinoma and their correlation with clinicopathological features

Farhana Ikram^a, Poorvi Mathur^a, Pallavi Dubey^{a,*}, Saloni Verma^b, Sanjay Agarwal^a, Shambhavi Tripathi^a

^a Department of Pathology, Hind Institute of Medical Sciences, Ataria, Sitapur road, Lucknow, Uttar Pradesh, India

^b King George's Medical University, Lucknow, Uttar Pradesh-26003, India

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ABSTRACT

Background: Oral squamous cell carcinoma (OSCC) is a leading cause of cancer-related morbidity and mortality worldwide, largely due to late-stage diagnosis and lack of reliable non-invasive biomarkers. Long non-coding RNAs (lncRNAs) such as HOTAIR, ANRIL, and MEG3 are emerging as potential liquid biopsy biomarkers in solid tumors. Current study evaluates the plasma expression levels of HOTAIR, ANRIL, and MEG3 in OSCC patients and assess their diagnostic potential and correlation with clinicopathological parameters.

Material and methods: In this case-control study, plasma samples were collected from 80 histologically confirmed treatment-naïve OSCC patients and 40 age- and sex-matched healthy controls. Quantitative real-time PCR (qRT-PCR) was performed to analyze lncRNA expression. Statistical analyses were conducted to assess differential expression, clinicopathological associations, and diagnostic performance via ROC curve analysis.

Results: Plasma HOTAIR and ANRIL were significantly upregulated (3.91- and 5.86-fold), while MEG3 was downregulated (0.12-fold) in OSCC compared to controls ($p < 0.001$ for all). Elevated levels of HOTAIR and ANRIL were associated with poor histological grade, higher T and N stage, LVI, PNI, necrosis, and advanced stage ($p = 0.001$). MEG3 levels decreased progressively with disease stage. Individually, each lncRNA achieved an AUC of 0.99 with high sensitivity and specificity. The three-lncRNA panel (HOTAIR + ANRIL + MEG3) yielded an AUC of 0.95, with 91.25 % sensitivity and 92.50 % specificity. MEG3 showed the best diagnostic performance for early-stage OSCC.

Conclusion: Plasma lncRNAs HOTAIR, ANRIL, and MEG3 show strong potential as non-invasive diagnostic biomarkers for OSCC, correlating with tumor aggressiveness and early disease detection. These findings support further validation in larger multicentric studies.

1. Introduction

Head and neck squamous cell carcinoma (HNSCC) ranks as the seventh most prevalent malignancy worldwide, with oral squamous cell carcinoma (OSCC) constituting the most common subtype, accounting for more than 40 % of HNSCC cases globally [1,2]. Despite advances in therapy, the five-year survival rate for OSCC has remained relatively stagnant, primarily due to late diagnosis and lack of effective early detection tools [3,4]. Early-stage identification remains critical for improving clinical outcomes, yet current diagnostic approaches are largely invasive, relying on tissue biopsies, and are insufficient for

screening or monitoring disease progression [5–7].

Histopathological examination remains the gold standard for OSCC diagnosis, utilizing hematoxylin and eosin (H&E) staining to assess cellular architecture and confirm malignancy. In poorly differentiated cases, immunohistochemical (IHC) markers such as p40, CK5/6, and pan-cytokeratin are employed for accurate tumor subtyping [8,9]. Radiological imaging, particularly contrast-enhanced magnetic resonance imaging (MRI) and contrast tomography (CT), plays a critical adjunctive role in evaluating tumor extent, bone involvement, and regional lymph node metastasis. Imaging findings not only assist in staging but also guide surgical planning and radiotherapy protocols [10,

* Corresponding author. Department of Pathology, Hind Institute of Medical Sciences, Lucknow, 225003, Uttar Pradesh, India.

E-mail address: pallavidubey002@gmail.com (P. Dubey).

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Established molecular and protein biomarkers have also been studied extensively for diagnostic, prognostic, and predictive purposes in OSCC. Overexpression of epidermal growth factor receptor (*EGFR*) and *cyclin D1*, *TP53* mutations, loss of *P16^{INK4A}*, and human papilloma virus (HPV) status have shown correlation with tumor progression and poor clinical outcomes [12–14]. *EGFR* overexpression is associated with aggressive tumor behaviour and resistance to therapy, while *cyclin D1* amplification is linked to nodal metastasis. *P16^{INK4A}* under expression is frequently observed in HPV-negative tumors and correlates with poorer prognosis [13,15]. However, the utility of these biomarkers is restricted to tissue-based evaluation and cannot support early or non-invasive detection.

In recent years, long noncoding RNAs (lncRNAs) have emerged as promising non-invasive biomarkers for OSCC. These non-protein-coding RNA molecules (>200 nucleotides) regulate gene expression at transcriptional, post-transcriptional, and epigenetic levels. Dysregulated lncRNA expression has been implicated in key oncogenic processes, including epithelial–mesenchymal transition (EMT), angiogenesis, apoptosis, immune evasion, and metastasis [16,17]. In OSCC tissue studies, lncRNAs such as *HOTAIR* (*HOX transcript antisense intergenic RNA*), *ANRIL* (*Antisense non-coding RNA in the INK4 locus*), and *MEG3* (*maternally expressed gene 3*) have demonstrated differential expression and functional effects on tumor biology [18–20]. Specifically, *HOTAIR* and *ANRIL* act as oncogenic lncRNAs, whereas *MEG3* functions as a tumor suppressor [21]. Among lncRNAs with established roles in OSCC tissues, elevated *HOTAIR* expression correlates with greater invasion and lymph node metastasis in multiple studies [22,23]. In contrast, *MEG3* expression is downregulated in OSCC tissues and cell lines, suppressing cell proliferation, migration, and invasion in vitro and correlating with higher clinical stage, nodal metastasis, and poor prognosis [24,25]. Overexpression of *MEG3* has been shown to increase *p53* levels, trigger apoptosis via inhibition of the Wnt/ β -catenin and JAK/STAT pathways, and selectively suppress aggressive phenotypes [25]. Similarly, *ANRIL* overexpression has been linked to enhanced proliferation and inhibited apoptosis via the *p15/p16* and Wnt/ β -catenin pathways, aligning with increased tumor aggressiveness [26,27].

Liquid biopsy studies indicate that circulating lncRNAs can serve as promising non-invasive diagnostic and prognostic markers in OSCC [28]. Dong Y et al., 2019, found low plasma levels of *CASC2* were associated with recurrence, while *PAPAS*, *UCA1*, *HOXA11-AS*, *LINC00964*, *MALAT1*, and *NCK1-AS1* were upregulated in OSCC patients and showed significant diagnostic potential with *PAPAS* differentiating early-stage OSCC and predicting worse survival [29,30]. Plasma *GAS5* predicted poor treatment response, though plasma levels of *HOTAIR* and lncRNA-p21 lacked conclusive results [31]. Nakamura K et al. identified a panel of four novel circulating lncRNAs that significantly discriminated early and late OSCC, underscored by ROC analyses [32]. Studies analysing lncRNAs in saliva or oral rinse are limited but promising; *MALAT1* and *HOTAIR* detected in saliva have shown differential expression in OSCC vs. healthy controls, correlating with disease stage and invasiveness [33]. In liquid biopsies, plasma lncRNAs have also been evaluated for their ability to monitor treatment response. Collectively, these findings underscore the potential of lncRNAs as diagnostic, prognostic, and therapeutic biomarkers in OSCC. However, despite strong evidence of *HOTAIR*, *ANRIL*, and *MEG3* dysregulation in OSCC tissues, and the promising roles of various lncRNAs in circulation, very few studies have evaluated plasma levels of *HOTAIR*, *ANRIL*, and *MEG3* in OSCC patients, and assessed their combined diagnostic capability or correlation with clinicopathological factors. The present study addresses this gap by quantifying plasma expression of *HOTAIR*, *ANRIL*, and *MEG3* in OSCC patients and comparing them to healthy controls. We further aim to correlate their plasma levels with clinicopathological parameters such as tumor stage, nodal involvement, and histological grade and evaluate their diagnostic performance for detecting early-stage disease. This investigation aims to elucidate the potential of

these lncRNAs as minimally invasive markers for disease aggressiveness, early diagnosis, in OSCC.

2. Material and methods

2.1. Study design and participant recruitment

This case-control study was conducted in the respective parents departments. The cases and controls were recruited from out patients department (OPD) of the institute, India. The testing of the samples was outsourced and performed at Matrix Biolab, Lucknow. A total of 120 participants were enrolled, comprising 80 histopathologically confirmed treatment-naïve OSCC patients and 40 age and sex-matched healthy controls. Patients eligible for inclusion in the study were those with histopathologically OSCC, diagnosed OSCC, with or without the aid of immunohistochemistry (IHC). Only treatment-naïve patients who provided written informed consent were included. Patients were excluded if their biopsy results indicated non-OSCC oral pathologies, if prior biopsy tissues were unavailable for review, or if they declined to participate in the study. Blood samples were collected before the initiation of any therapy. Demographic and clinicopathological data were recorded using a structured case record form. The study protocol was reviewed and approved by the Institutional Ethics Committee of the institute (Approval number 639/Ethics/2023 dated 26/06/2023). Written informed consent was obtained from all participants prior to enrolment.

2.2. Blood collection and plasma preparation

Peripheral venous blood (5.0 mL) was collected preoperatively from OSCC cases and healthy controls into EDTA vacutainers (BD Vacutainer, USA). Plasma was separated by a two-step centrifugation process: initially at 1900×g for 10 min, followed by 16,000×g for 10 min at room temperature to ensure removal of cellular debris. The plasma aliquots were immediately stored at −80 °C until further processing for RNA extraction.

3. Diagnostic confirmation (Gold standard)

All enrolled cases underwent comprehensive diagnostic evaluation, which served as the gold standard for confirmation of OSCC and staging was done as per the AJCC [34]. Clinical assessment involved detailed documentation of patient demographics, clinical presentation, tobacco use history, lesion site and laterality, and surgical details. Histopathological evaluation was conducted on H&E-stained sections from FFPE blocks to confirm OSCC morphology, and assess for features such as tumor differentiation and necrosis. In cases with ambiguous or poorly differentiated histological features, a diagnostic immunohistochemistry panel including *p40* (Clone: BD28, BioCare, USA), *CK5/6* (Clone: D5/16 B4, DAKO, Denmark), and *Pan-Cytokeratin* (Clone: AE1/AE3, DAKO, Denmark) was employed to confirm squamous differentiation and OSCC diagnosis.

4. RNA isolation and quantitative real-time PCR

Total RNA was extracted from plasma samples using TRIzol LS reagent (Invitrogen, Carlsbad, CA, USA), following the manufacturer's protocol. RNA yield and quality were assessed, and 500 ng of RNA was reverse-transcribed to complementary DNA (cDNA) using the Revert Aid H Minus First Strand cDNA Synthesis Kit (Thermo Fisher Scientific, USA). The synthesized cDNA was stored at −20 °C for downstream applications. Quantitative real-time PCR (qRT-PCR) was conducted using SYBR Green PCR Master Mix (Applied Biosystems, USA) on the CFX96 Real-Time PCR Detection System (Bio-Rad, USA) as presented in Fig. 1. The reactions were set in a final volume of 20 μ L, containing 10 μ L SYBR Green mix, 1 μ L each of forward and reverse primers, 2 μ L of cDNA, and

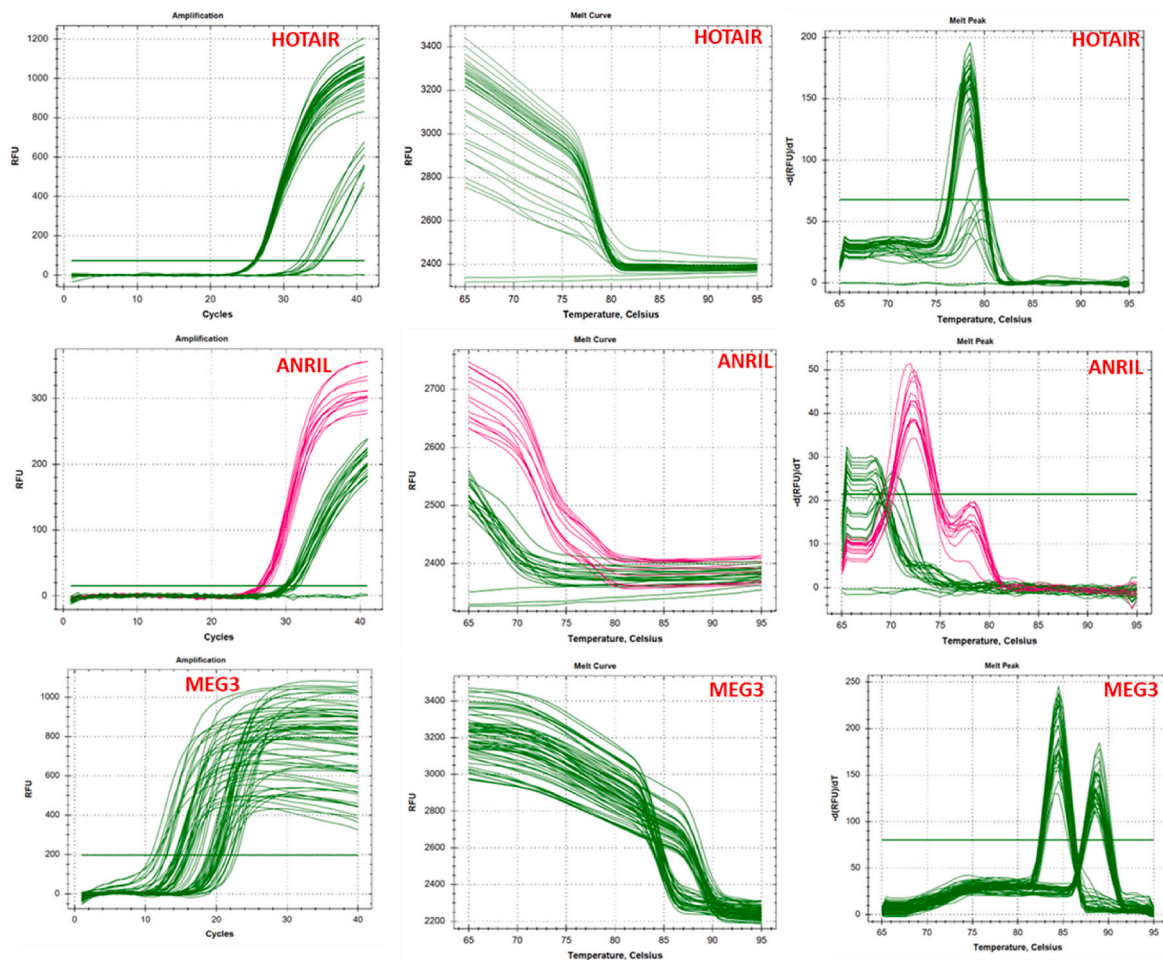


Fig. 1. Amplification, melt curve, and melt peak analysis of plasma lncRNAs HOTAIR, ANRIL, and MEG3 using qRT-PCR. Representative plots show (left to right) the amplification curves, melt curves, and melt peak profiles for the long non-coding RNAs HOTAIR (top row), ANRIL (middle row), and MEG3 (bottom row). Amplification curves indicate exponential fluorescence increase confirming successful target amplification. Melt curve and melt peak analyses demonstrate the specificity of each qRT-PCR product.

7 μ L nuclease-free water. Primer sequences and annealing temperatures for the studied lncRNAs (*HOTAIR*, *ANRIL*, *MEG3*) are listed in [Supplementary Table 1](#). Thermal cycling conditions included: initial denaturation at 95 °C for 10 min, followed by 35 cycles of denaturation at 95 °C for 30 s, annealing (as per [Supplementary Table 1](#)) for 30 s, and extension at 72 °C for 30 s. A melt curve analysis was performed from 65 °C to 95 °C with 0.1 °C increments to confirm amplification specificity. GAPDH was used as an internal control for normalization. Relative quantification of lncRNA expression was determined using the $2^{-\Delta\Delta CT}$ method. The ΔCT value was calculated by subtracting the CT of GAPDH from the CT of the target lncRNA. $\Delta\Delta CT$ was obtained by subtracting the average ΔCT of controls from each case's ΔCT . The fold change in expression was expressed as $2^{\Delta\Delta CT}$. A higher mean relative fold change in OSCC cases compared to controls indicated upregulation, whereas a lower fold change suggested downregulation.

5. Statistical analysis

Categorical variables were summarized as frequencies and percentages, while continuous variables were expressed as mean $\pm$ standard error (SE). To evaluate the association between the relative expression levels of lncRNAs and various demographic and clinicopathological characteristics, appropriate statistical tests were applied: the Mann-Whitney *U* test for comparisons between two groups, and either one-way ANOVA or the Kruskal-Wallis test for comparisons involving

more than two groups, depending on data distribution. The diagnostic performance of each lncRNA was assessed using receiver operating characteristic (ROC) curve analysis. Sensitivity, specificity, and predictive values (positive and negative) were computed using MedCalc software. A p-value of less than 0.05 was considered statistically significant. All statistical analyses were conducted using GraphPad Prism version 9 (GraphPad Software, San Diego, CA, USA).

6. Results

A total of 80 histopathologically confirmed OSCC cases were included in this prospective case-control study. Among the participants, 50.0 % ($n = 40$) were aged ≤ 40 years and 50.0 % ($n = 40$) were > 40 years. The majority of the cases were male (75.0 %, $n = 60$), with a history of tobacco use in 63.8 % ($n = 51$) and alcohol consumption in 81.3 % ($n = 65$). The most common tumor sites were the tongue (38.8 %, $n = 31$) and buccal mucosa/gingivobuccal sulcus (37.5 %, $n = 30$), followed by retromolar trigone (8.8 %), hard palate (6.2 %), alveolus (5.0 %), and lip (3.7 %). Lesions were almost equally distributed laterally, with 47.5 % on the right and 52.5 % on the left. Histopathologically, moderately differentiated (MD) tumors predominated (52.5 %, $n = 42$), followed by well-differentiated (WD) tumors (37.5 %, $n = 30$) and poorly differentiated (PD) tumors (10.0 %, $n = 8$). The primary tumor size (T-stage) distribution showed 23.8 % with T1, 25.0 % with T2, 26.2 % with T3, and 25.0 % with T4 tumors. Nodal involvement was

evident in 60.0 % of cases: N1 and N2 in 22.5 % each, and N3 in 15.0 %. Lymphovascular invasion (LVI) and perineural invasion (PNI) were observed in 50.0 % (n = 40) and 47.5 % (n = 38) of the cases, respectively. Tumor necrosis was present in 58.8 % (n = 47). Based on AJCC pathological staging, 18.8 % were in Stage I, 18.8 % in Stage II, 26.2 % in Stage III, and 36.2 % in Stage IV.

7. Expression of plasma lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* in OSCC vs. controls

The comparative analysis of plasma lncRNA expression levels between OSCC patients and healthy controls revealed statistically significant alterations across all three studied markers (Table 1, Fig. 2). Plasma levels of *HOTAIR* were significantly upregulated in OSCC patients (mean $\pm$ SD: 4.27 ± 1.48) compared to controls (1.09 ± 0.38), with a 3.91-fold increase ($p < 0.0001$). Similarly, *ANRIL* expression was markedly higher in OSCC cases (5.91 ± 1.68) than in controls (1.01 ± 0.29), showing a 5.86-fold upregulation ($p < 0.0001$). Conversely, *MEG3* expression was significantly downregulated in OSCC patients (0.14 ± 0.11) relative to the control group (1.16 ± 0.60), indicating an 88 % decrease ($p < 0.0001$).

8. Association of plasma lncRNAs with demographic characteristics in OSCC patients

The association of circulating plasma lncRNA levels with demographic parameters among OSCC patients is summarized in Table 2. There were no statistically significant differences in plasma levels of *HOTAIR*, *ANRIL*, and *MEG3* with respect to age group (≤ 40 vs. > 40 years), gender, or tobacco use (all $p > 0.05$). Although plasma *HOTAIR* levels were higher in alcohol consumers (4.26 ± 1.67) compared to non-consumers (3.30 ± 1.39), this difference was not statistically significant ($p = 0.087$). However, *ANRIL* levels were significantly elevated among alcohol users (5.00 ± 1.88) compared to non-users (4.16 ± 1.59), with a statistically significant association ($p = 0.017$). When analyzed by tumor site, significant variation in expression levels of all three lncRNAs was observed. Patients with buccal mucosa/gingivobuccal sulcus tumors exhibited the highest mean plasma *HOTAIR* (4.65 ± 1.78), *ANRIL* (5.42 ± 1.98), and *MEG3* (0.15 ± 0.09) levels compared to those with lesions at the tongue, alveolus, palate, or lip. These differences were statistically significant ($p = 0.003$, $p < 0.001$, and $p = 0.008$ for *HOTAIR*, *ANRIL*, and *MEG3* respectively). No significant association was observed between tumor laterality (right vs. left) and the expression levels of any of the studied lncRNAs (all $p > 0.05$).

9. Association of plasma lncRNAs with clinicopathological features in OSCC

Plasma expression levels of *HOTAIR*, *ANRIL*, and *MEG3* were evaluated in relation to clinicopathological characteristics of OSCC patients (Table 3). Histological grade was significantly associated with all three lncRNAs. Patients with MD tumors exhibited significantly higher *HOTAIR* expression (4.75 ± 1.58) than WD (3.25 ± 1.24) and PD tumors (3.40 ± 1.27) ($p < 0.001$). Similarly, *ANRIL* expression was significantly elevated in PD tumors (7.15 ± 1.34) compared to WD (4.87 ± 2.05) and MD (5.00 ± 1.70) grades ($p = 0.003$). *MEG3* expression also differed significantly by grade ($p = 0.001$), with the lowest expression in MD cases (0.17 ± 0.14) and highest in PD (0.25 ± 0.23). Significant

associations were also observed between tumor stage (T-stage) and *HOTAIR*, *ANRIL*, and *MEG3* expression. *HOTAIR* levels increased progressively with T-stage, from 2.53 ± 0.79 in T1 to 5.54 ± 1.49 in T4 tumors ($p = 0.001$). A similar trend was seen with *ANRIL* ($p = 0.002$) and *MEG3* ($p = 0.002$). Nodal status (N-stage) was positively associated with *HOTAIR* ($p < 0.001$), *ANRIL* ($p = 0.003$), and *MEG3* ($p = 0.006$). Notably, *HOTAIR* expression rose from 3.26 ± 1.19 in N0 to 5.61 ± 1.54 in N3 disease. *MEG3* levels were highest in N3 stage (0.22 ± 0.19). Presence of LVI was significantly correlated with elevated *HOTAIR* (4.93 ± 1.57 vs. 3.23 ± 1.23 ; $p = 0.001$), *ANRIL* (5.28 ± 1.63 vs. 4.56 ± 1.95 ; $p = 0.001$), and *MEG3* (0.18 ± 0.12 vs. 0.16 ± 0.18 ; $p = 0.001$). PNI was also significantly associated with higher *HOTAIR* ($p = 0.001$) and *ANRIL* ($p = 0.046$), though no significant difference was observed in *MEG3* expression between PNI-positive and -negative groups ($p = 0.145$). Presence of tumor necrosis was associated with increased levels of *HOTAIR* (4.71 ± 1.65 vs. 3.26 ± 1.24 ; $p = 0.001$), *ANRIL* ($p = 0.001$), and *MEG3* ($p = 0.002$). Finally, a significant stepwise rise in *HOTAIR* levels was noted across pathological stages, from 2.37 ± 0.63 in Stage I to 5.15 ± 1.55 in Stage IV ($p < 0.001$). *ANRIL* also increased significantly with advancing stage ($p = 0.001$), and *MEG3* followed a similar trend ($p = 0.003$).

10. Stage-wise expression of plasma lncRNAs in OSCC

To further explore the relationship between plasma lncRNAs and disease severity, their expression levels were compared across OSCC stages and normal controls (Table 4, Fig. 3). A statistically significant stepwise increase in *HOTAIR* levels was observed from early-stage (Stage I + II: 3.08 ± 0.12) to late-stage disease (Stage III + IV: 5.09 ± 0.21), compared to healthy controls (1.09 ± 0.38). The difference was significant when comparing controls with Stage I + II ($p = 0.012$), controls with Stage III + IV ($p < 0.0001$), and Stage I + II with Stage III + IV ($p = 0.023$), indicating progressive upregulation of *HOTAIR* with advancing tumor stage. *ANRIL* expression followed a similar pattern, with marked elevation in both stage I + II (5.78 ± 0.11) and stage III + IV (5.61 ± 0.18) relative to controls (1.01 ± 0.29). These differences were statistically significant between controls and stage I + II ($p = 0.015$) and controls and stage III + IV ($p = 0.0002$), although the difference between stage I + II and stage III + IV was not significant ($p = 0.331$), suggesting early and sustained elevation of *ANRIL* in OSCC. Conversely, *MEG3* expression was significantly downregulated in OSCC. Its levels dropped from 1.16 ± 0.60 in controls to 0.17 ± 0.03 in stage I + II and 0.15 ± 0.02 in stage III + IV. These reductions were significant in both comparisons to controls (stage I + II: $p = 0.021$; stage III + IV: $p = 0.0003$), while the difference between early and late stages was not statistically significant ($p = 0.681$), suggesting that *MEG3* down-regulation occurs early in disease progression.

11. Diagnostic performance of plasma lncRNAs for OSCC detection

Receiver operating characteristic (ROC) curve analysis was performed to evaluate the diagnostic utility of plasma lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* in distinguishing OSCC patients from healthy controls (Table 5, Fig. 4). Plasma *HOTAIR* demonstrated excellent diagnostic ability with an AUC of 0.99 (95 % CI: 0.98–1.00, $p = 0.001$). At a cutoff value of ≥ 1.70 , it achieved 100 % sensitivity (95 % CI: 90.0–100.0 %) and 95 % specificity (95 % CI: 90.0–100.0 %), with a positive predictive

Table 1

Comparison of plasma lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* levels between OSCC and normal controls.

Marker	OSCC Mean $\pm$ SD	Control Mean $\pm$ SD	p value	% fold change compared to control	Change compared to control
Plasma <i>HOTAIR</i>	4.27 ± 1.48	1.09 ± 0.38	<0.0001	3.91	Upregulated
Plasma <i>ANRIL</i>	5.91 ± 1.68	1.01 ± 0.29	<0.0001	5.86	Upregulated
Plasma <i>MEG3</i>	0.14 ± 0.11	1.16 ± 0.60	<0.0001	0.12	Downregulated

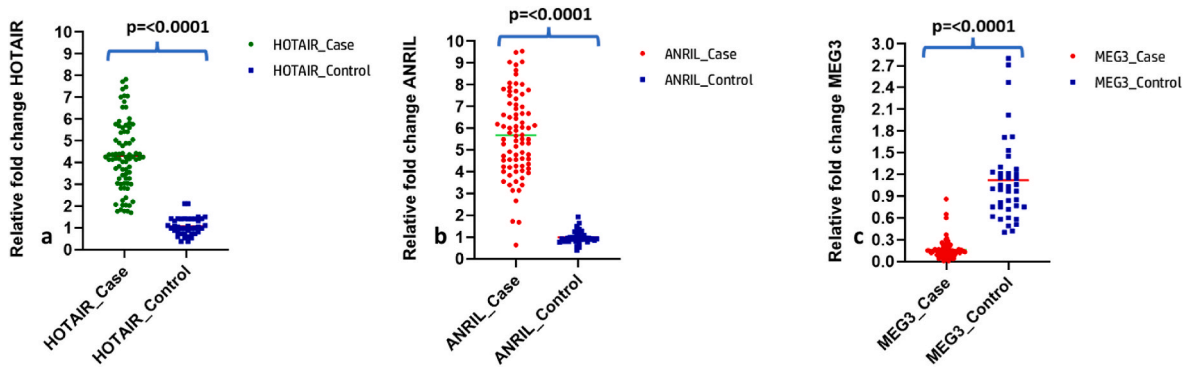


Fig. 2. Scatter Dot plots showing the relative expression of (A) HOTAIR, (B) ANRIL, and (C) MEG3 in plasma samples from OSCC patients ($n = 80$) and healthy controls ($n = 40$), as determined by qRT-PCR. Each dot represents an individual value. Mean values are indicated by horizontal lines. HOTAIR and ANRIL were significantly upregulated, while MEG3 was significantly downregulated in OSCC patients compared to controls ($p < 0.001$, Mann–Whitney U test for all comparisons). These findings highlight the diagnostic potential of these lncRNAs as circulating biomarkers in OSCC.

Table 2
Association of Plasma lncRNA with Demographic Characteristics of OSCC.

Demographic Characteristics	n (%)	HOTAIR (Mean $\pm$ SD)	P value	ANRIL (Mean $\pm$ SD)	P value	MEG3 (Mean $\pm$ SD)	P value
Age group							
≤ 40	40 (50.0 %)	4.01 $\pm$ 1.50	0.861	4.65 $\pm$ 1.52	0.498	0.16 $\pm$ 0.10	0.735
> 40	40 (50.0 %)	4.15 $\pm$ 1.75		5.19 $\pm$ 2.02		0.18 $\pm$ 0.18	
Gender							
Male	60 (75.0 %)	4.13 $\pm$ 1.65	0.679	4.88 $\pm$ 1.84	0.527	0.17 $\pm$ 0.16	0.581
Female	20 (25.0 %)	3.92 $\pm$ 1.58		4.94 $\pm$ 1.80		0.18 $\pm$ 0.14	
Tobacco intake							
Yes	51 (63.8 %)	4.32 $\pm$ 1.71	0.228	5.15 $\pm$ 1.93	0.070	0.19 $\pm$ 0.18	0.130
No	29 (36.2 %)	3.70 $\pm$ 1.45		4.46 $\pm$ 1.66		0.14 $\pm$ 0.09	
Alcohol							
Yes	65 (81.3 %)	4.26 $\pm$ 1.67	0.087	5.00 $\pm$ 1.88	0.017	0.17 $\pm$ 0.15	0.388
No	15 (18.7 %)	3.30 $\pm$ 1.39		4.16 $\pm$ 1.59		0.19 $\pm$ 0.18	
Site of tumor							
Buccal mucosa/GBS	30 (37.5 %)	4.65 $\pm$ 1.78	0.003	5.42 $\pm$ 1.98	< 0.001	0.15 $\pm$ 0.09	0.008
Tongue	31 (38.8 %)	3.71 $\pm$ 1.40		4.77 $\pm$ 1.90		0.16 $\pm$ 0.13	
Retromolar trigone	7 (8.8 %)	4.78 $\pm$ 1.74		5.47 $\pm$ 1.39		0.22 $\pm$ 0.18	
Alveolus	4 (5.0 %)	4.42 $\pm$ 1.73		6.91 $\pm$ 1.62		0.29 $\pm$ 0.30	
Hard Palate	5 (6.2 %)	4.14 $\pm$ 0.65		4.98 $\pm$ 1.64		0.25 $\pm$ 0.23	
Lip	3 (3.7 %)	4.24 $\pm$ 2.70		7.16 $\pm$ 1.73		0.18 $\pm$ 0.06	
Laterality							
Right	38 (47.5 %)	4.50 $\pm$ 1.79	0.053	5.05 $\pm$ 1.89	0.512	0.17 $\pm$ 0.14	0.866
Left	42 (52.5 %)	3.69 $\pm$ 1.40		4.78 $\pm$ 1.84		0.17 $\pm$ 0.16	

value (PPV) of 97.6 % and a negative predictive value (NPV) of 100 %, yielding an overall diagnostic accuracy of 98.3 %. ANRIL showed similarly robust performance, with an AUC of 0.99 (95 % CI: 0.96–1.00, $p = 0.001$). Using a cutoff of ≥ 2.66 , it achieved 96.3 % sensitivity, 100 % specificity, and both PPV and NPV of 100 % and 93.0 %, respectively, with an accuracy of 97.5 %. MEG3, a tumor-suppressive lncRNA, was significantly downregulated in OSCC, and at a cutoff of ≤ 0.37 , showed an AUC of 0.99 (95 % CI: 0.98–1.00, $p = 0.001$). It achieved 96.3 % sensitivity, 100 % specificity, with PPV and NPV of 100 % and 93.0 %, respectively, and an accuracy of 97.5 %.

The combination of markers further improved diagnostic precision. The three-lncRNA panel (HOTAIR + ANRIL + MEG3) yielded the highest combined AUC of 0.95 (95 % CI: 0.92–0.98), with 91.25 % sensitivity, 92.50 % specificity, 96.7 % PPV, and 82.1 % NPV, with an overall accuracy of 91.7 %. Among two-lncRNA panels, HOTAIR + MEG3 showed superior performance (AUC = 0.94) compared to HOTAIR + ANRIL (AUC = 0.93) and ANRIL + MEG3 (AUC = 0.92).

12. Diagnostic value of plasma lncRNAs in early-stage OSCC (stage I + II)

To assess the potential of plasma lncRNAs as early diagnostic biomarkers, ROC curve analysis was performed comparing patients with

early-stage OSCC (Stage I + II) to healthy controls (Table 5). HOTAIR demonstrated a strong diagnostic performance with an AUC of 0.89 (95 % CI: 0.83–0.96, $p < 0.001$). At a cutoff value of ≥ 2.48 , HOTAIR showed a sensitivity of 83.87 % (95 % CI: 67.4–93.0 %) and specificity of 82.50 % (95 % CI: 67.2–92.0 %), with PPV and NPV of 81.2 % and 84.8 %, respectively, yielding an overall diagnostic accuracy of 83.10 %. ANRIL also showed significant diagnostic potential with an AUC of 0.86 (95 % CI: 0.78–0.93, $p < 0.001$). Using a cutoff of ≥ 3.75 , ANRIL achieved a sensitivity of 80.65 % (95 % CI: 63.7–91.2 %) and specificity of 80.00 % (95 % CI: 64.1–90.3 %), with PPV of 78.1 %, NPV of 82.1 %, and accuracy of 80.28 %. Notably, MEG3 exhibited the highest diagnostic accuracy among the three markers in early-stage OSCC. With a cutoff of ≤ 0.18 , it yielded an AUC of 0.91 (95 % CI: 0.86–0.97, $p < 0.001$). MEG3 achieved a sensitivity of 87.10 % (95 % CI: 71.2–95.3 %) and specificity of 85.00 % (95 % CI: 70.2–93.5 %), with PPV of 84.4 %, NPV of 87.5 %, and an overall accuracy of 85.92 %.

13. Discussion

OSCC remains a major global health challenge, especially in regions such as South and Southeast Asia where tobacco, betel nut, and alcohol consumption are highly prevalent. Despite significant advances in surgery, radiotherapy, and systemic therapy, the overall 5-year survival

Table 3
Association of Plasma lncRNA with Clinicopathological Characteristics.

Clinicopathological Characteristics	n (%)	HOTAIR (Mean ± SD)	p-value	ANRIL (Mean ± SD)	p-value	MEG3 (Mean ± SD)	p-value
Histological Grade							
(WD)	30 (37.5 %)	3.25 ± 1.24	<0.001	4.87 ± 2.05	0.003	0.16 ± 0.13	0.001
(MD)	42 (52.5 %)	4.75 ± 1.58		5.00 ± 1.70		0.17 ± 0.14	
(PD)	8 (10.0 %)	3.40 ± 1.27		7.15 ± 1.34		0.25 ± 0.23	
T-Stage							
T1	19 (23.8 %)	2.53 ± 0.79	0.001	4.95 ± 1.80	0.002	0.19 ± 0.22	0.002
T2	20 (25.0 %)	3.61 ± 1.20		4.75 ± 1.93		0.16 ± 0.07	
T3	21 (26.2 %)	4.40 ± 1.18		4.74 ± 1.55		0.14 ± 0.06	
T4	20 (25.0 %)	5.54 ± 1.49		5.66 ± 1.89		0.20 ± 0.15	
N-Stage							
N0	32 (40.0 %)	3.26 ± 1.19	<0.001	4.73 ± 1.95	0.003	0.16 ± 0.13	0.006
N1	18 (22.5 %)	4.35 ± 1.20		4.76 ± 1.57		0.15 ± 0.06	
N2	18 (22.5 %)	5.07 ± 1.54		5.69 ± 1.90		0.17 ± 0.10	
N3	12 (15.0 %)	5.61 ± 1.54		5.73 ± 1.61		0.22 ± 0.19	
LVI							
Evident	40 (50.0 %)	4.93 ± 1.57	0.001	5.28 ± 1.63	0.001	0.18 ± 0.12	0.001
Not Evident	40 (50.0 %)	3.23 ± 1.23		4.56 ± 1.95		0.16 ± 0.18	
PNI							
Evident	38 (47.5 %)	4.71 ± 1.63	0.001	5.09 ± 1.72	0.046	0.17 ± 0.10	0.145
Not Evident	42 (52.5 %)	3.50 ± 1.44		4.77 ± 1.93		0.17 ± 0.18	
Tumor Necrosis							
Present	47 (58.8 %)	4.71 ± 1.65	0.001	5.21 ± 1.76	0.001	0.18 ± 0.13	0.002
Absent	33 (41.2 %)	3.26 ± 1.24		4.49 ± 1.90		0.15 ± 0.17	
Pathological Staging							
Stage I	15 (18.8 %)	2.37 ± 0.63	<0.001	4.81 ± 2.07	0.001	0.20 ± 0.24	0.003
Stage II	15 (18.8 %)	3.61 ± 1.20		4.75 ± 1.93		0.16 ± 0.07	
Stage III	21 (26.2 %)	4.40 ± 1.18		4.74 ± 1.55		0.14 ± 0.06	
Stage IV	29 (36.2 %)	5.15 ± 1.55		5.53 ± 1.76		0.19 ± 0.14	

Table 4
Comparison of relative expression of miRNA in serum sample of normal control (NC) and stage of OSCC.

Relative expression in plasma (Mean ± SE)				p value comparing		
Gene	Control (n = 40)	stage I + II (n = 30)	stage III + IV (n = 50)	control with stage II + III	Control with stage III + IV	stage II + III with stage III + IV
<i>HOTAIR</i>	1.09 ± 0.38	3.08 ± 0.12	5.09 ± 0.21	0.012	0.0001	0.023
<i>ANRIL</i>	1.01 ± 0.29	5.78 ± 0.11	5.61 ± 0.18	0.015	0.0002	0.331
<i>MEG3</i>	1.16 ± 0.60	0.17 ± 0.03	0.15 ± 0.02	0.021	0.0003	0.681

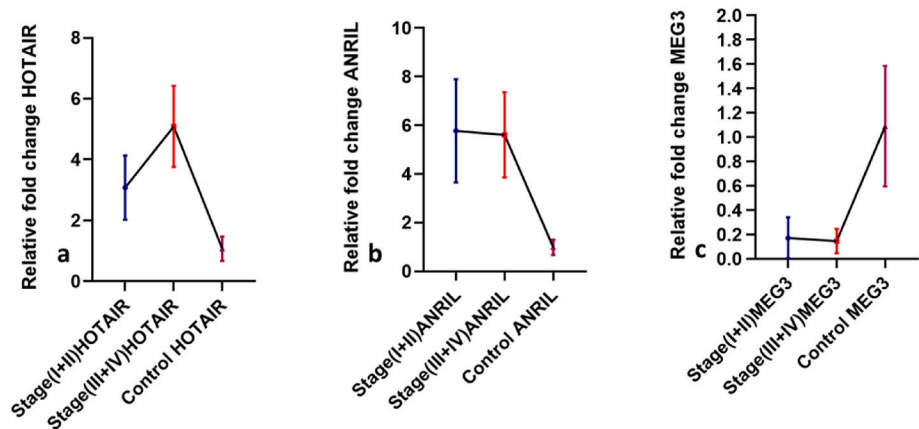


Fig. 3. Stage-wise comparison of plasma lncRNA expression levels in OSCC patients and healthy controls. Bar graphs display the mean of relative plasma expression levels of (a) HOTAIR, (b) ANRIL, and (c) MEG3 among healthy controls and OSCC patients stratified by pathological stage: early-stage (Stage I + II) and late-stage (Stage III + IV). A progressive increase in HOTAIR levels and a sustained elevation of ANRIL expression were observed across tumor stages. MEG3 expression was significantly downregulated in both early and late-stage OSCC compared to controls. All lncRNA comparisons between OSCC groups and controls were statistically significant ($p < 0.05$, one-way ANOVA with post hoc analysis), highlighting their potential role in tumor staging and early detection.

rate for OSCC has stagnated around 50–60 %, largely due to late-stage diagnosis and the lack of effective non-invasive screening tools. The identification of robust, blood-based biomarkers capable of detecting OSCC at early stages and predicting prognosis is, therefore, a clinical priority. In the present study, we investigated the plasma levels of

HOTAIR, *ANRIL*, and *MEG3* in OSCC patients and healthy controls. Our findings revealed significant dysregulation of these lncRNAs in OSCC patients, with *HOTAIR* and *ANRIL* being markedly upregulated, and *MEG3* significantly downregulated, compared to controls. These alterations were not only statistically significant but also biologically

Table 5
Diagnostic values of plasma lncRNA in OSCC.

OSCC vs. Normal Control								
Marker	Cut off value	AUC (95 % CI)	p-value	Sensitivity (95 % CI)	Specificity (95 % CI)	PPV (95 % CI)	NPV (95 % CI)	Accuracy (95 % CI)
<i>HOTAIR</i>	≥1.70	0.99 (0.98–1.00)	0.001	100 % (90.0–100.0)	95 % (90.0–100.0)	97.6 % (95.2–100.0)	100 % (83.3–100.0)	98.3 % (93.3–100.0)
<i>ANRIL</i>	≥2.66	0.99 (0.96–1.00)	0.001	96.3 % (93.8–100.0)	100 % (97.5–100.0)	100 % (98.7–100.0)	93.0 % (88.9–100.0)	97.5 % (95.8–100.0)
<i>MEG3</i>	≤0.37	0.99 (0.98–1.00)	0.001	96.3 % (91.3–100.0)	100 % (100.0–100.0)	100 % (100.0–100.0)	93.0 % (85.1–100.0)	97.5 % (94.2–100.0)
Combination Diagnostic								
<i>HOTAIR</i> + <i>ANRIL</i>	3.42	0.93 (0.90–0.97)	<0.001	87.50 % (78.5–93.2 %)	90.00 % (76.3–96.6 %)	94.6 % (87.3–97.9 %)	78.3 % (64.0–88.0 %)	88.3 % (81.6–93.0 %)
<i>HOTAIR</i> + <i>MEG3</i>	1.89	0.94 (0.91–0.97)	<0.001	90.00 % (81.5–95.0 %)	87.50 % (73.2–95.0 %)	93.8 % (86.4–97.3 %)	81.4 % (67.4–90.3 %)	89.2 % (82.6–93.6 %)
<i>ANRIL</i> + <i>MEG3</i>	2.22	0.92 (0.88–0.96)	<0.001	85.00 % (75.6–91.3 %)	85.00 % (70.2–93.5 %)	91.9 % (83.6–96.3 %)	74.5 % (60.1–85.0 %)	85.0 % (77.9–90.3 %)
<i>HOTAIR</i> + <i>ANRIL</i> + <i>MEG3</i>	2.61	0.95 (0.92–0.98)	<0.001	91.25 % (83.0–95.8 %)	92.50 % (79.6–97.8 %)	96.7 % (90.3–99.1 %)	82.1 % (68.3–91.0 %)	91.7 % (85.9–95.4 %)
Early stage (I + II) vs. Normal control								
<i>HOTAIR</i>	2.48	0.89 (0.83–0.96)	<0.001	83.87 % (67.4–93.0 %)	82.50 % (67.2–92.0 %)	81.2 % (65.4–91.1 %)	84.8 % (69.1–93.5 %)	83.10 % (73.3–90.0 %)
<i>ANRIL</i>	3.75	0.86 (0.78–0.93)	<0.001	80.65 % (63.7–91.2 %)	80.00 % (64.1–90.3 %)	78.1 % (61.2–89.0 %)	82.1 % (66.5–91.7 %)	80.28 % (70.2–87.8 %)
<i>MEG3</i>	0.18	0.91 (0.86–0.97)	<0.001	87.10 % (71.2–95.3 %)	85.00 % (70.2–93.5 %)	84.4 % (68.3–93.5 %)	87.5 % (72.6–95.3 %)	85.92 % (76.7–92.1 %)

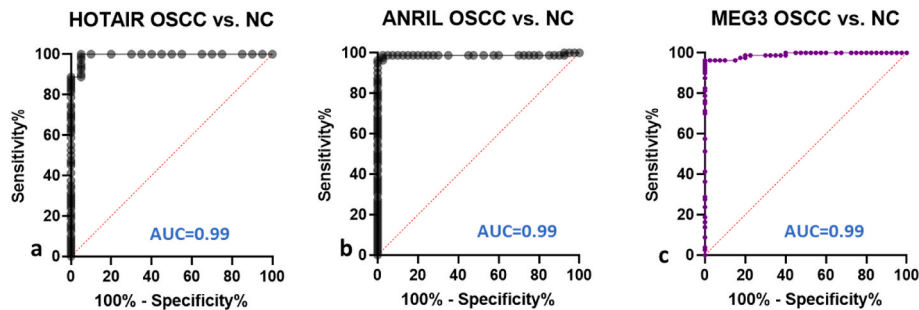


Fig. 4. Receiver operating characteristic (ROC) curve analysis for plasma lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* in distinguishing OSCC from healthy controls. ROC curves demonstrate the diagnostic performance of (A) *HOTAIR*, (B) *ANRIL*, and (C) *MEG3* plasma expression levels. All three lncRNAs exhibited excellent discriminative ability with area under the curve (AUC) values of 0.99, indicating high sensitivity and specificity for OSCC detection. The dashed red diagonal line represents the line of no discrimination (AUC = 0.50). These results underscore the strong diagnostic potential of circulating lncRNAs as non-invasive biomarkers for OSCC.

plausible, as these lncRNAs have been previously linked with tumor progression, epigenetic regulation, and cell cycle control in epithelial malignancies. To our knowledge, this is among the few studies to demonstrate the simultaneous diagnostic evaluation of these three specific lncRNAs in plasma of OSCC patients.

Our study identified significant dysregulation of circulating lncRNAs in OSCC patients, with *HOTAIR* and *ANRIL* markedly upregulated (3.91- and 5.86-fold increases, respectively), while *MEG3* was substantially downregulated (0.12-fold decrease) relative to healthy controls. These findings align closely with tissue-based data, where elevated *HOTAIR* and *ANRIL* expression correlated with tumor aggressiveness, invasion, and metastasis in OSCC and other squamous cell carcinomas [35]. For instance, *HOTAIR* has been implicated in promoting EMT through PRC2-mediated histone modifications, correlating with poor prognosis and lymph node metastasis in head and neck cancers [24,36]. Similarly, *ANRIL* overexpression has been linked to enhanced cellular proliferation and apoptotic resistance through dysregulation of the *p15/p16* and Wnt/ β -catenin pathways [37]. Though previous research mainly focused on tumor tissues, our plasma-based findings extend these observations, indicating systemic release and potential utility in non-invasive diagnostics.

The profound downregulation of *MEG3* is consistent with its known tumor-suppressive functions in OSCC. Studies have demonstrated that *MEG3* suppresses proliferation, migration, and invasion, in part by sponging oncogenic miR-21 and miR-548d-3p, influencing JAK/STAT and Wnt/ β -catenin pathways [36]. *MEG3* repressive impact on β -catenin and promotion of apoptosis via p53 signaling have also been documented in OSCC cell lines [26]. Notably, our plasma measurements reflect these tissue-level changes, supporting *MEG3* viability as a circulating biomarker. Together, these mechanistic insights support our hypothesis that elevated plasma *HOTAIR* and *ANRIL* levels signify aggressive, proliferative tumor behaviour, while decreased *MEG3* reflects impaired tumor suppressive regulation. This altered lncRNA expression profile in plasma may mirror intratumoral biology and offers promise as a minimally invasive diagnostic and prognostic tool in OSCC.

In our study, we evaluated the relationship between circulating plasma levels of lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* with demographic characteristics of OSCC. We observed no statistically significant difference in lncRNA expression between age groups, suggesting that expression of these non-coding RNAs in plasma is likely governed by tumor biology rather than patient age. Similarly, gender did not significantly influence expression levels of any of the three lncRNAs, in

agreement with prior findings that gender has limited impact on circulating lncRNA profiles in head and neck cancers [38]. Interestingly, while tobacco exposure a known etiological factor for OSCC did not show a statistically significant association with lncRNA levels, a trend towards higher *HOTAIR* and *ANRIL* expression was seen among tobacco users. More notably, *ANRIL* expression was significantly higher in alcohol users ($p = 0.017$), suggesting a potential synergistic role of alcohol in modulating *ANRIL* transcription or release into circulation. Previous studies have indicated that alcohol can alter chromatin remodeling and lncRNA expression through oxidative stress and epigenetic deregulation, which may in part explain our findings [39,40].

Among all demographic variables, tumor site showed a statistically significant association with the expression of *HOTAIR* ($p = 0.003$), *ANRIL* ($p < 0.001$), and *MEG3* ($p = 0.008$). Buccal mucosa, gingivo-buccal sulcus (GBS), and retromolar trigone tumors exhibited the highest mean expression levels of *HOTAIR* and *ANRIL*, whereas *MEG3* expression was highest in tumors arising from the alveolus and hard palate. These findings suggest a possible site-specific variation in lncRNA dynamics, which may reflect differences in local tumor micro-environments, exposure to carcinogens, or anatomical patterns of lymphatic spread and immune interaction. Such subsite-specific differences in molecular profiles have been reported previously in head and neck squamous cell carcinoma and may have implications for personalized diagnostic and therapeutic strategies [41,42]. Lastly, tumor laterality did not show any significant difference in lncRNA levels, indicating that lncRNA expression is not influenced by tumor side but more likely by intrinsic molecular behaviour and aggressiveness. Overall, our data reinforce that plasma lncRNAs such as *HOTAIR*, *ANRIL*, and *MEG3* are reflective of tumor biological characteristics rather than demographic variables, further underscoring their potential role as non-invasive biomarkers for disease detection and subsite-specific evaluation in OSCC.

In the present study, we observed that plasma levels of lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* were significantly associated with various adverse clinicopathological features of OSCC. Notably, expression of *HOTAIR* and *ANRIL* increased significantly with higher histological grade, T and N stage, LVI, PNI, tumor necrosis, and advanced pathological stage. *MEG3* also showed a rising trend in plasma levels across several aggressive tumor parameters, particularly higher histological grade, T and N stage, tumor necrosis, and advanced staging. These findings are in line with the known biological functions of these lncRNAs. *HOTAIR* has been widely studied in multiple cancers, including OSCC, and is known to promote tumor progression by interacting with the polycomb repressive complex 2 (PRC2) and silencing tumor suppressor genes such as E-cadherin [22,43]. Its overexpression has been correlated with EMT, invasion, and metastasis in HNSCC, as well as poor overall survival [22,36,44]. Similarly, *ANRIL* promotes oncogenesis by epigenetically repressing tumor suppressor genes like *CDKN2A/B* and has been reported to facilitate proliferation, migration, and invasion in OSCC tissues [45,46]. Our observation of elevated *ANRIL* levels in plasma samples from patients with high-grade, high-stage tumors further supports its role as an oncogenic lncRNA and a potential non-invasive biomarker.

In contrast to its classically defined tumor-suppressor role in various cancers through activation of *p53* and inhibition of angiogenesis and proliferation, *MEG3* showed a somewhat paradoxical increase in plasma levels in more advanced OSCC cases in our study. This may be attributed to tumor necrosis or apoptotic shedding of intracellular contents, releasing *MEG3* into circulation. While most studies report decreased *MEG3* expression in tumor tissues, similar discrepancies between tissue and plasma levels have been observed in other malignancies, suggesting a complex interplay between intracellular regulation and extracellular release dynamics [47–49].

Collectively, the correlation of circulating lncRNAs with pathological parameters such as T and N stage, tumor grade, LVI, PNI, and tumor necrosis highlights their potential utility as non-invasive biomarkers.

Unlike tissue-based assays, plasma lncRNA profiling offers a dynamic, repeatable, and clinically feasible approach to assessing tumor biology in real time. The significant differential expression of *HOTAIR*, *ANRIL*, and *MEG3* in aggressive tumor subgroups suggests their role in stratifying OSCC patients based on risk, aiding early diagnosis, prognosis, and treatment monitoring. These findings add to a growing body of evidence supporting the use of circulating lncRNAs in the clinical management of OSCC and warrant further validation in larger, prospective cohorts. These findings suggest that *MEG3* holds particular promise as a single biomarker for early OSCC detection, outperforming other circulating RNAs previously evaluated. In contrast, while *HOTAIR* and *ANRIL* also demonstrate robust diagnostic potential in early-stage disease, *MEG3* superior AUC and balanced sensitivity–specificity profile supports its prioritization in plasma-based screening strategies. This enhanced accuracy could significantly advance early diagnostic pathways, especially in high-risk populations where early detection dramatically improves survival outcomes.

Our findings demonstrate that circulating lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* hold significant diagnostic value in differentiating OSCC from healthy controls. All three individual biomarkers showed remarkably high AUC values of 0.99 (95 % CI: 0.96–1.00) with p -values < 0.001 , indicating outstanding diagnostic performance. *HOTAIR*, with a cutoff of ≥ 1.70 , achieved 100 % sensitivity and 95 % specificity, while *ANRIL* (cutoff ≥ 2.66) and *MEG3* (cutoff ≤ 0.37) each reached 100 % specificity with sensitivities of 96.3 %. These results suggest that aberrant plasma expression of these lncRNAs not only occurs in OSCC patients but does so with clear discriminatory power, which is consistent with prior reports highlighting the stability and disease-specific release of circulating lncRNAs [50–53]. The diagnostic potential of *MEG3* is particularly noteworthy given its role as a tumor suppressor; its downregulation in OSCC aligns with studies in other solid tumors where *MEG3* loss has been linked to proliferation and poor prognosis [54]. Conversely, the upregulation of *HOTAIR* and *ANRIL* corroborates previous studies showing their oncogenic roles in cell migration, invasion, and epigenetic regulation in various cancers, including HNSCC [53,55–57].

Furthermore, combination analysis of lncRNAs enhanced diagnostic efficiency. The triple-panel of *HOTAIR* + *ANRIL* + *MEG3* yielded an AUC of 0.95, with 91.25 % sensitivity and 92.50 % specificity, outperforming all dual-marker combinations. This synergistic diagnostic gain supports the growing evidence that multiplexed lncRNA signatures provide superior accuracy compared to single biomarkers [Shi T et al., 2016]. The diagnostic accuracy (91.7 %) and high predictive values (PPV: 96.7 %, NPV: 82.1 %) further underscore the robustness of this lncRNA panel in identifying OSCC. Taken together, our data highlight the clinical promise of plasma lncRNA profiling as a non-invasive tool for OSCC detection, particularly in resource-constrained settings where early diagnosis remains challenging. These biomarkers may complement conventional histopathological diagnosis and could serve as frontline screening tools or adjuncts in monitoring disease progression and therapeutic response.

The present study is limited by its single-center design, relatively small sample size, and lack of longitudinal follow-up to assess prognostic implications of lncRNA expression. Additionally, functional validation of *HOTAIR*, *ANRIL*, and *MEG3* in oral tumorigenesis was not performed. Future studies should include multicenter cohorts with larger and more diverse populations to validate diagnostic utility. Integration of longitudinal data could establish these lncRNAs as prognostic or predictive markers. Functional assays and mechanistic studies are warranted to elucidate the biological roles of these lncRNAs in OSCC progression, which may further support their use as therapeutic targets in personalized medicine.

14. Conclusion

This study highlights the significant diagnostic potential of circulating lncRNAs *HOTAIR*, *ANRIL*, and *MEG3* in oral squamous cell

carcinoma. Their altered plasma expression patterns are associated with distinct clinicopathological features, suggesting their involvement in tumor progression and aggressiveness. The findings support the role of these lncRNAs as minimally invasive biomarkers for early detection and disease monitoring. Importantly, their combined diagnostic performance further enhances sensitivity and specificity, emphasizing their clinical applicability. These results pave the way for future translational research exploring lncRNA-based diagnostics and therapeutic interventions, with potential to improve personalized management and outcomes in patients with oral squamous cell carcinoma.

Ethical approval/patient consent

Study was approved by the institutional ethics committee of the institute. A written informed consent was obtained from each participant.

CRediT authorship contribution statement

Farhana Ikram, Poorvi Mathur, Pallavi Dubey, Saloni Verma: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, writing original draft, Writing review & editing, Visualization, Project administration, Funding acquisition. **Sanjay Agarwal:** Methodology, Resources, Formal analysis, Writing review & editing. **Pallavi Dubey, Shambhavi Tripathi:** Supervision, Resources, Validation, Writing review & editing.

Ethical approval/patient consent

This study was approved by the institutional ethics committee of the institute (Approval number 639/Ethics/2023 dated 26/06/2023). Written informed consent was obtained from all patients.

Declaration of generative AI in scientific writing

AI tool (ChatGPT) was used to support language editing. All scientific content was created and validated by the authors.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jlb.2025.100315>.

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